

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

08

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

The table shows selected values from function f .

x	fx
-1	16
0	17
1	18
2	19

Which of the following is the best description of function f ?

- A. Decreasing linear
- B. Increasing linear
- C. Decreasing exponential
- D. Increasing exponential

Q2

PROBLEM-SOLVING AND DATA ANALYSIS

An object travels at a constant speed of 12 centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel 108 centimeters?

- A. 9
- B. 96
- C. 120
- D. 972

Q3

PROBLEM-SOLVING AND DATA ANALYSIS

A waiter receives tips from each customer. On average, the tip is 15% of the customer's bill. At this rate, which of the following is closest to the tip the waiter can expect when a customer has a bill that is \$78.20?

- A. \$8.00
- B. \$10.00
- C. \$12.00
- D. \$14.00

Customer Purchases at a Gas Station

	Beverage purchased	Beverage not purchased	Total
Gasoline purchased	60	25	85
Gasoline not purchased	35	15	50
Total	90	40	135

On Tuesday, a local gas station had 135 customers. The table above summarizes whether or not the customers on Tuesday purchased gasoline, a beverage, both, or neither. Based on the data in the table, what is the probability that a gas station customer selected at random on that day did not purchase gasoline?

- A. $\frac{15}{50}$
- B. $\frac{15}{40}$
- C. $\frac{35}{50}$
- D. $\frac{50}{135}$

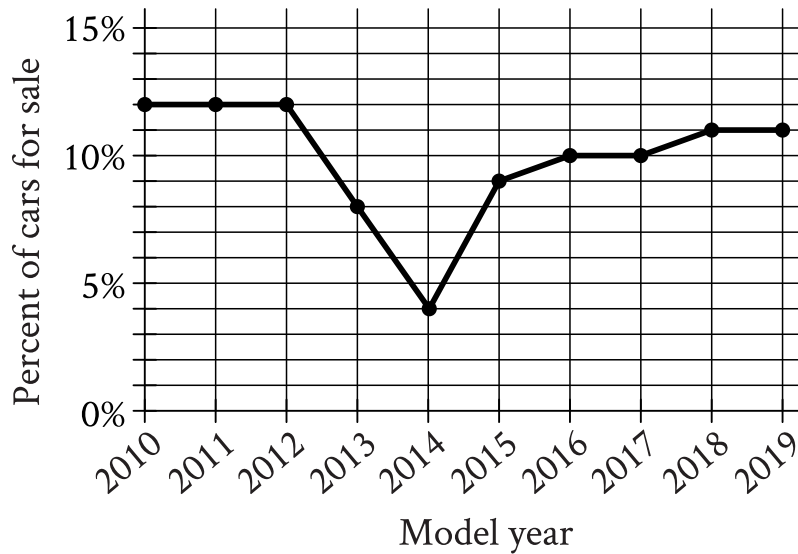
Data set A: 1, 2, 3, 4, 5, 6, 7

Data set B: 1, 1, 2, 2, 3, 3, 4

Which of the following statements correctly compares the means of data set A and data set B?

- A. The mean of each data set is 2.
- B. The mean of each data set is 4.
- C. The mean of data set A is less than the mean of data set B.
- D. The mean of data set A is greater than the mean of data set B.

The line graph shows the percent of cars for sale at a used car lot on a given day by model year.



- The line graph:
 - Begins at 2010, 12%
 - Remains level to 2011, 12%
 - Remains level to 2012, 12%
 - Falls sharply to 2013, 8%
 - Falls sharply to 2014, 4%
 - Rises sharply to 2015, 9%
 - Rises gradually to 2016, 10%
 - Remains level to 2017, 10%
 - Rises gradually to 2018, 11%
 - Remains level to 2019, 11%

For what model year is the percent of cars for sale the smallest?

A. 2012

B. 2013

C. 2014

D. 2015

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

An object's speed is 64 yards per second. What is the object's speed, in feet per second? 1 yard = 3 feet

A. 61

B. 67

C. 94

D. 192

Call Ratings					
	1 Star	2 Stars	3 Stars	4 Stars	Total
Employee A	16	49	72	8	145
Employee B	4	10	22	34	70
Employee C	8	56	45	16	125
Employee D	22	42	84	12	160
Total	50	157	223	70	500

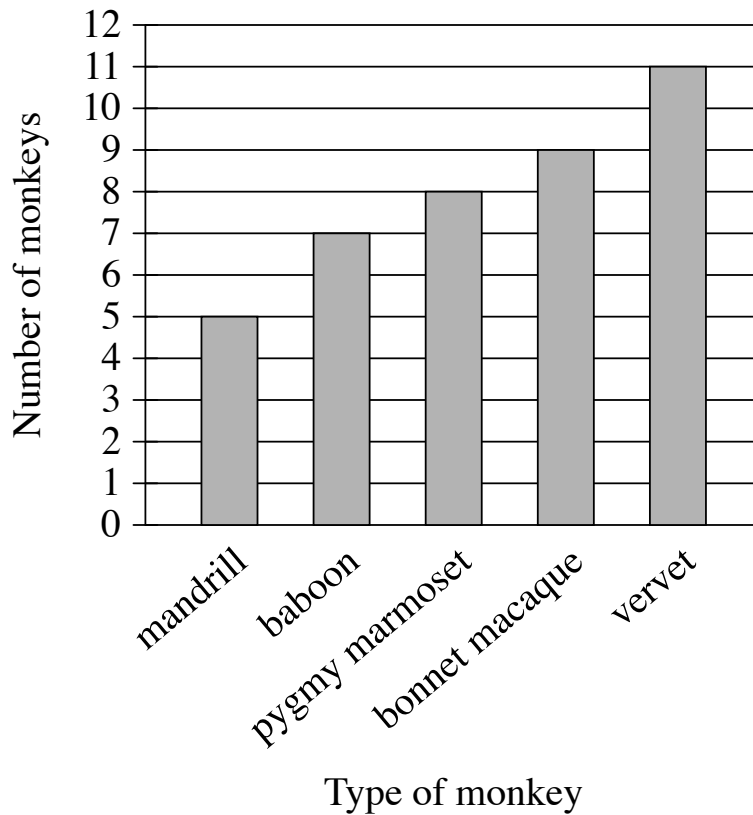
A supervisor at a call center reviewed 500 calls taken by four employees and rated the employees' performance on each call on a scale from 1 star to 4 stars. The ratings for each employee are shown in the table above. According to the table, to the nearest whole percent, what percent of Employee A's calls received a rating of 1 star?

- A. 3%
- B. 11%
- C. 16%
- D. 32%

A bag contains a total of 60 marbles. A marble is to be chosen at random from the bag. If the probability that a blue marble will be chosen is 0.35, how many marbles in the bag are blue?

- A. 21
- B. 25
- C. 35
- D. 39

The bar graph shows the number of each type of monkey in a sanctuary.



- The Number of monkeys data for the 5 bars are as follows:
 - mandrill: 5
 - baboon: 7
 - pygmy marmoset: 8
 - bonnet macaque: 9
 - vervet: 11

How many more vervets are in this sanctuary than mandrills?

A. 11

B. 6

C. 5

D. 3